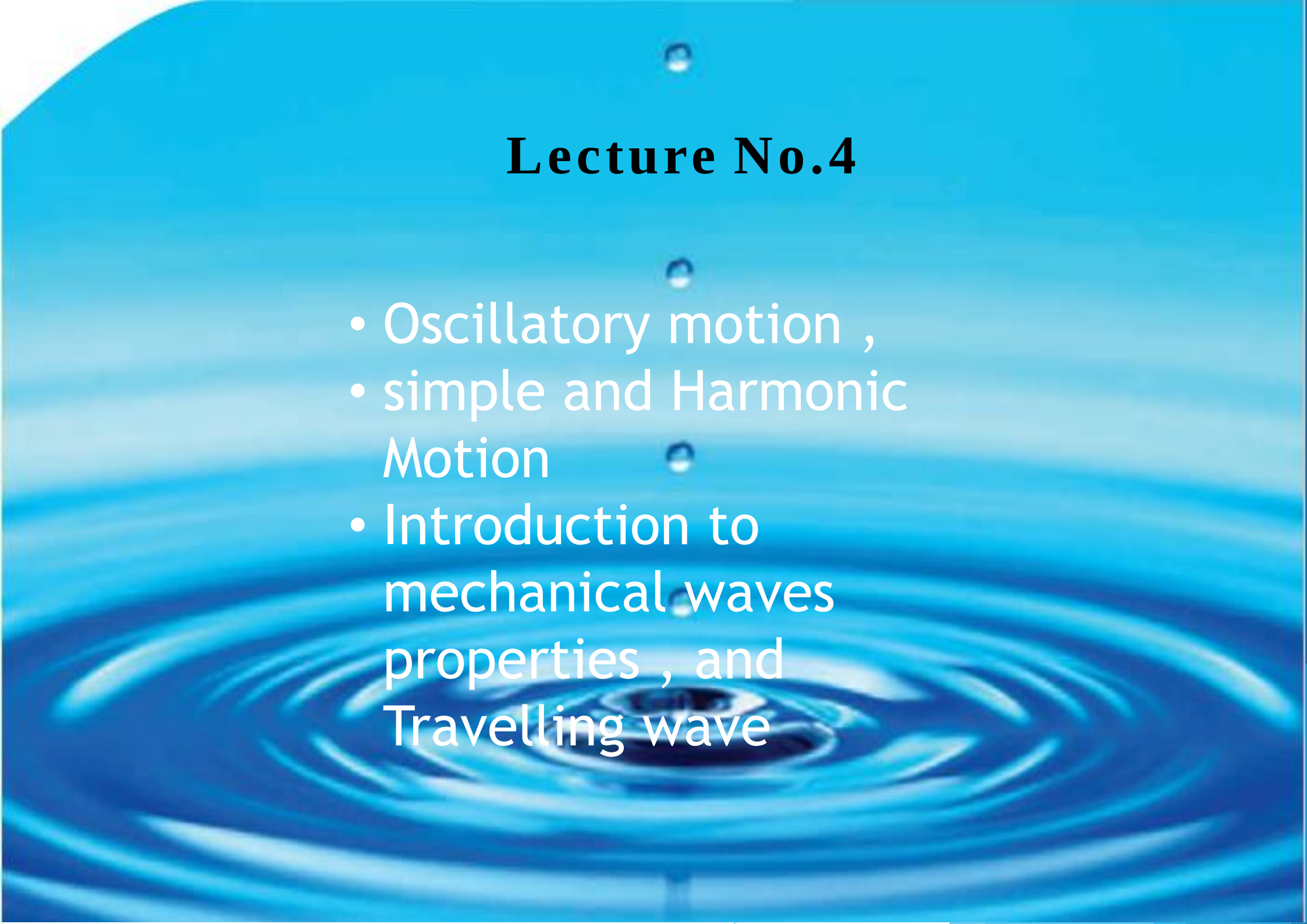
Physics I

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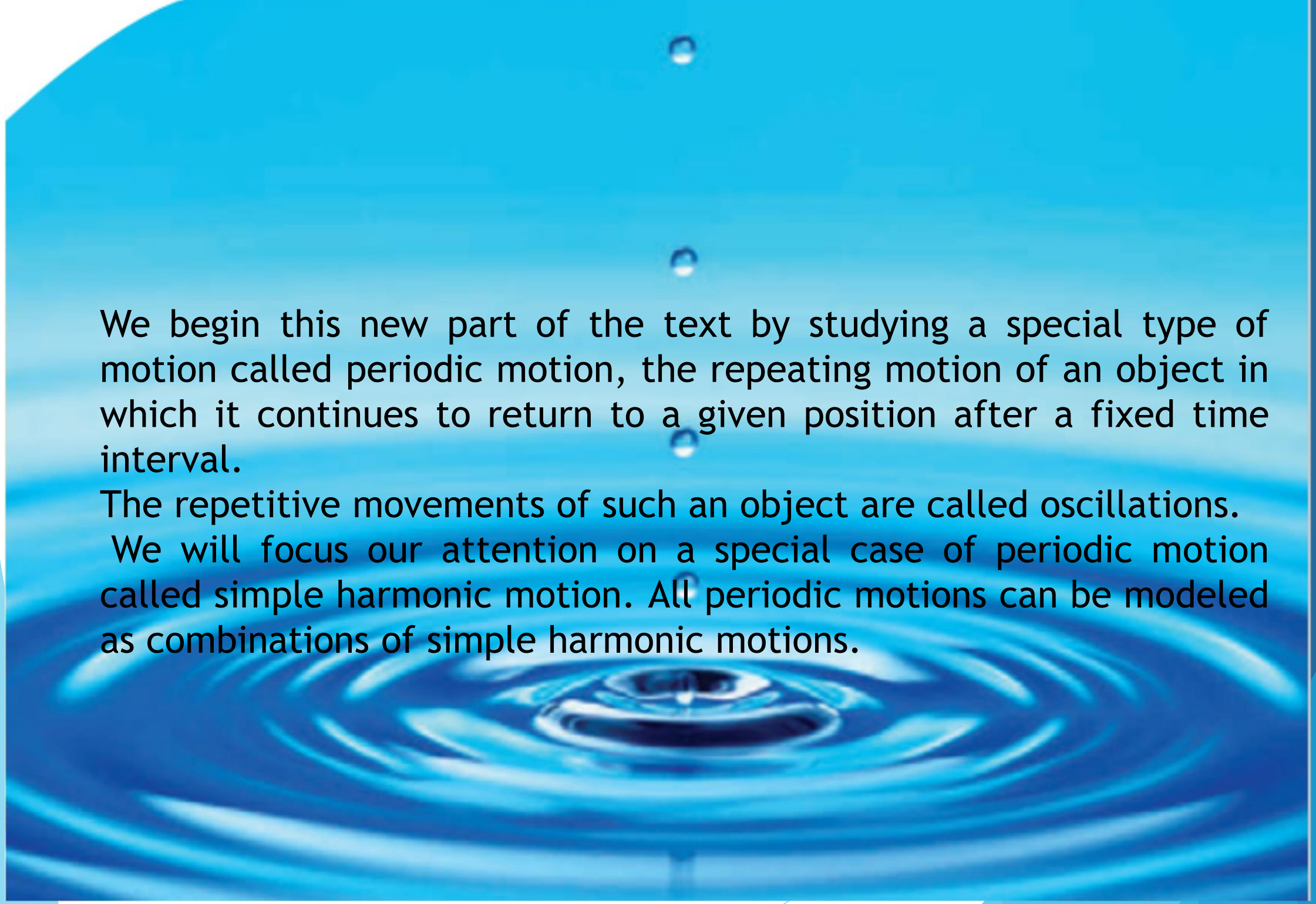
Dr. Mohamed anwar

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Lecture No.4

- Oscillatory motion ,
- simple and Harmonic Motion
- Introduction to mechanical waves properties , and Travelling wave



We begin this new part of the text by studying a special type of motion called periodic motion, the repeating motion of an object in which it continues to return to a given position after a fixed time interval.

The repetitive movements of such an object are called oscillations.

We will focus our attention on a special case of periodic motion called simple harmonic motion. All periodic motions can be modeled as combinations of simple harmonic motions.

Periodic motion

is motion of an object that regularly returns to a given position after a fixed time interval.

simple harmonic motion

A special kind of periodic motion occurs in mechanical systems when the force acting on an object is proportional to the position of the object relative to some equilibrium position and the force is always directed toward the equilibrium position

Motion of an Object Attached to a Spring

As a model for simple harmonic motion, consider a block of mass m attached to the end of a spring, with the block free to move on a frictionless, horizontal surface (Fig. 15.1). When the spring is neither stretched nor compressed, the block is at rest at the position called the **equilibrium position** of the system, which we identify as $x = 0$ (Fig. 15.1b). We know from experience that such a system oscillates back and forth if disturbed from its equilibrium position.

We can understand the oscillating motion of the block in Figure 15.1 qualitatively by first recalling that when the block is displaced to a position x, the spring exerts on the block a force that is proportional to the position and given by **Hooke's law**

$$F_s = -kx \tag{15.1}$$

We call F_s a **restoring force** because it is always directed toward the equilibrium position and therefore *opposite* the displacement of the block from equilibrium.

When the block is displaced from the equilibrium point and released, it is a particle under a net force and consequently undergoes an acceleration. Applying the particle under a net force model to the motion of the block, with Equation 15.1 providing the net force in the x direction, we obtain

$$\sum F_x = ma_x \rightarrow -kx = ma_x$$

$$a_x = -\frac{k}{m}x \tag{15.2}$$

That is, the acceleration of the block is proportional to its position, and the direction of the acceleration is opposite the direction of the displacement of the block from equilibrium. Systems that behave in this way are said to exhibit **simple harmonic motion.** An object moves with simple harmonic motion whenever its acceleration is proportional to its position and is oppositely directed to the displacement from equilibrium.

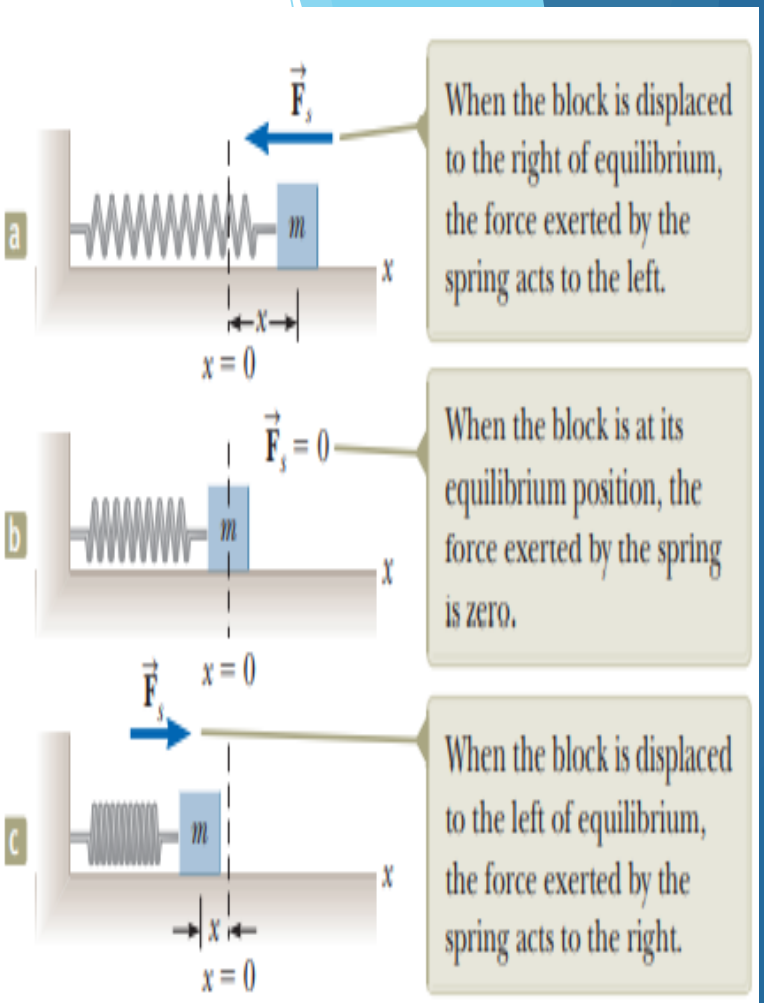


Figure 15.1 A block attached to a spring moving on a frictionless surface.

Particle in Simple Harmonic Motion

The motion described in the preceding section occurs so often that we identify the **particle in simple harmonic motion** model to represent such situations. To develop a mathematical representation for this model, we will generally choose x as the axis along which the oscillation occurs; hence, we will drop the subscript-x notation in this discussion. Recall that, by definition, $a = dv/dt = d^2x/dt^2$, so we can express Equation 15.2 as

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x \tag{15.3}$$

If we denote the ratio k/m with the symbol ω^2 , then

$$\omega^2 = \frac{k}{m} \tag{15.4}$$

and Equation 15.3 can be written in the form

$$\frac{d^2x}{dt^2} = -\omega^2 x \tag{15.5}$$

Let's now find a mathematical solution to Equation 15.5, that is, a function $x(t)$ that satisfies this second-order differential equation and is a mathematical representation of the position of the particle as a function of time. We seek a function whose second derivative is the same as the original function with a negative sign and multiplied by ω^2. The trigonometric functions sine and cosine exhibit this behavior, so we can build a solution around one or both of them. The following cosine function is a solution to the differential equation:

$$x(t) = A\cos(\omega t + \phi) \tag{15.6}$$

where A, v, and f are constants. To show explicitly that this solution satisfies Equation 15.5, notice that

$$\frac{dx}{dt} = A\frac{d}{dt}\cos(\omega t + \phi) = -\omega A \sin(\omega t + \phi) \qquad (15.7)$$

$$\frac{d^2x}{dt^2} = -\omega A\frac{d}{dt}\sin(\omega t + \phi) = -\omega^2 A\cos(\omega t + \phi) \qquad (15.8)$$

Comparing Equations 15.6 and 15.8, we see that $d^2x/dt^2 = -\omega^2 x$ and Equation 15.5 is satisfied.

The parameters A, ω, and Φ are constants of the motion.

a graphical representation of the motion by plotting x as a function of t is shown in Figure 15.2a.

- A, called the amplitude of the motion, is simply:

the maximum value of the position of the particle in either the positive or negative x direction

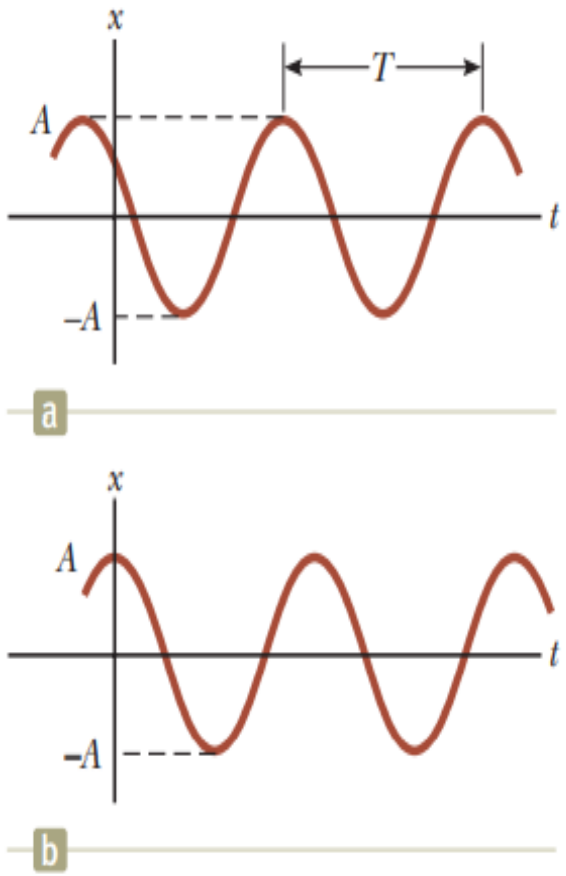


Figure 15.2 (a) An x–t graph for a particle undergoing simple harmonic motion. The amplitude of the motion is A, and the period (defined in Eq. 15.10) is T. (b) The x–t graph for the special case in which $x = A$ at $t = 0$ and hence $\phi = 0$.

ω is called the angular frequency, and it has units of radians per second. It is a measure of how rapidly the oscillations are occurring; the more oscillations per unit time, the higher the value of ω.

From Equation 15.4, the angular frequency is

$$\omega = \sqrt{\frac{k}{m}}$$

Φis called the phase constant which is:

The initial phase angle

The quantity (ωt + Φ) is called the phase of the motion

Notice :

the function x(t) is periodic and its value is the same each time ωt increases by 2π radians.

Equations 15.1, 15.5, and 15.6 form the basis of the mathematical representation of the particle in simple harmonic motion model.

- If you are analyzing a situation and find that the force on an object modeled as a particle is of the mathematical form of Equation 15.1, you know the motion is that of a simple harmonic oscillator and the position of the particle is described by Equation 15.6.
- If you analyze a system and find that it is described by a differential equation of the form of Equation 15.5, the motion is that of a simple harmonic oscillator.
- If you analyze a situation and find that the position of a particle is described by Equation 15.6, you know the particle undergoes simple harmonic motion.

Let us investigate further the mathematical description of simple harmonic motion.

The period T of the motion

is the time interval required for the particle to go through one full cycle of its motion

$$T = \frac{2\pi}{\omega}$$

the frequency *f* of the motion.

The inverse of the period is called the frequency *f and is defined as:*

the number of oscillations the particle undergoes per unit time interval

$$f = \frac{1}{T} = \frac{\omega}{2\pi}$$

The units of f are cycles per second, or hertz (Hz)

it can be express the period and frequency of the motion for the particle in simple harmonic motion in terms of the characteristics m and k of the system as

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

$$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

We can obtain the velocity and acceleration[2] of a particle undergoing simple harmonic motion from Equations 15.7 and 15.8:

$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi) \tag{15.15}$$

$$a = \frac{d^2x}{dt^2} = -\omega^2 A \cos(\omega t + \phi) \tag{15.16}$$

From Equation 15.15, we see that because the sine and cosine functions oscillate between ± 1, the extreme values of the velocity v are $\pm\omega A$. Likewise, Equation 15.16 shows that the extreme values of the acceleration a are $\pm\omega^2 A$. Therefore, the *maximum* values of the magnitudes of the velocity and acceleration are

$$v_{max} = \omega A = \sqrt{\frac{k}{m}}\,A \tag{15.17}$$

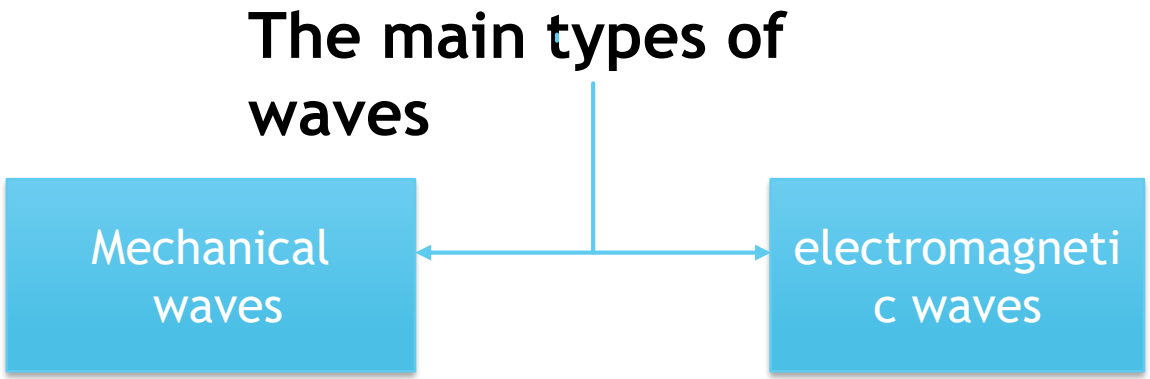
$$a_{max} = \omega^2 A = \frac{k}{m}A \tag{15.18}$$

Energy of the Simple Harmonic Oscillator

Total energy of a simple harmonic oscillator

$$E = \frac{1}{2}kA^2$$

Introduction to mechanical waves



1. . In the case of mechanical waves, some physical medium is being disturbed; in our pebble example, elements of water are disturbed.
2. All mechanical waves require (1) some source of disturbance, (2) a medium containing elements that can be disturbed, and (3) some physical mechanism through which elements of the medium can influence each other
3. Electromagnetic waves do not require a medium to propagate; some examples of electromagnetic waves are visible light, radio waves, television signals, and x-rays

Propagation of a Disturbance

- As the pulse in Figure.1 travels, each disturbed element of the string moves in a direction perpendicular to the direction of propagation.
- A traveling wave or pulse that causes the elements of the disturbed medium to move perpendicular to the direction of propagation is called a transverse wave.

a transverse wave

A traveling wave or pulse that causes the elements of the disturbed medium to move perpendicular to the direction of propagation

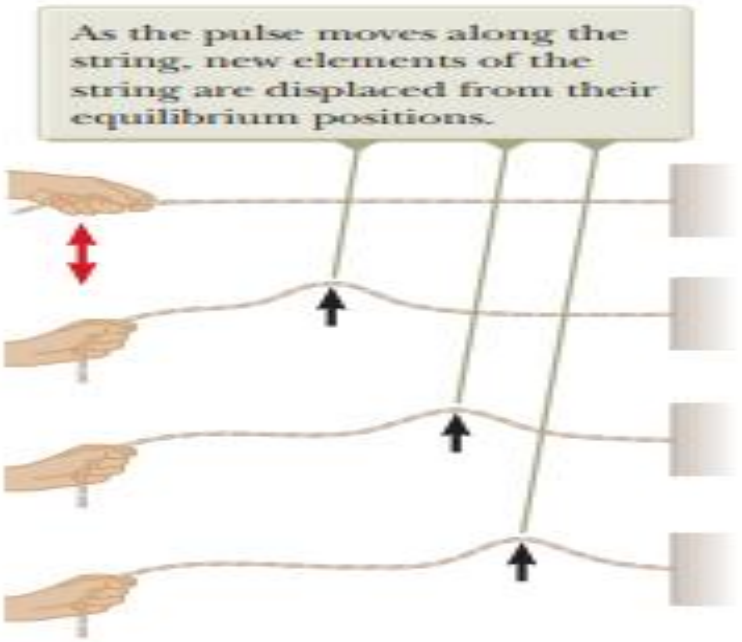


Figure .1 A hand moves the end of a stretched string up and down once (red arrow), causing a pulse to travel along the string.

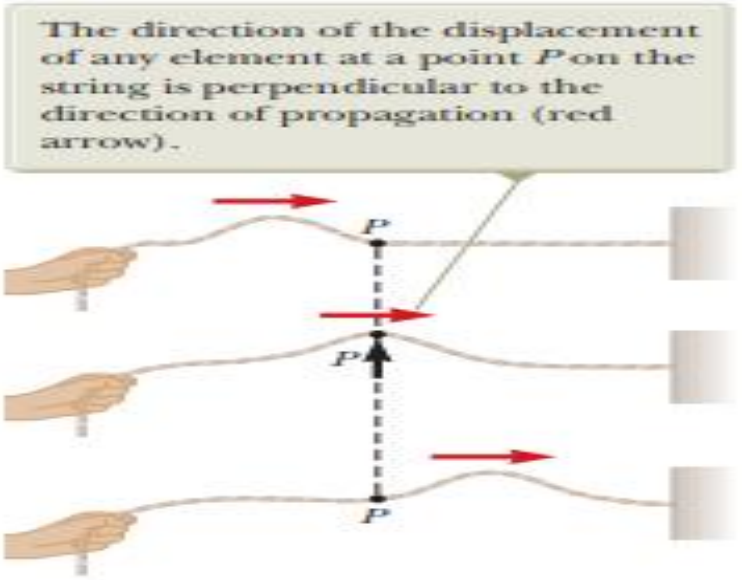


Figure .2 The displacement of a particular string element for a transverse pulse traveling on a stretched string.

- Compare this wave with another type of pulse, one moving down a long, stretched spring as shown in Figure.3. The left end of the spring is pushed briefly to the right and then pulled briefly to the left. This movement creates a sudden compression of a region of the coils. The compressed region travels along the spring (to the right in Fig..3).
- Notice that the direction of the displacement of the coils is parallel to the direction of propagation of the compressed region.
- A traveling wave or pulse that causes the elements of the medium to move parallel to the direction of propagation is called a longitudinal wave.
- Sound waves, are another example of longitudinal waves. The disturbance in a sound wave is a series of high-pressure and

low-pressure regions that travel through air.

a longitudinal wave

A traveling wave or pulse that causes the elements of the medium to move parallel to the direction of propagation

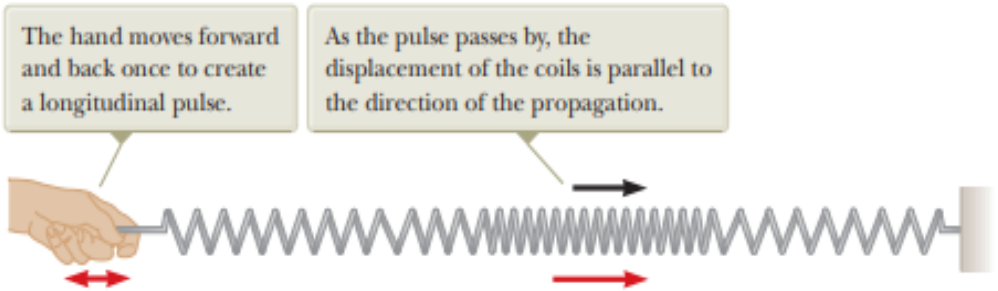


Figure .3 A longitudinal pulse along a stretched spring.

[1]In reality, the pulse changes shape and gradually spreads out during the motion. This effect, called *dispersion*, is common to many mechanical waves as well as to electromagnetic waves. We do not consider dispersion in this chapter.

Consider a pulse traveling to the right on a long string as shown in Figure .5.

In general, then, we can represent the transverse position y for all positions and times, measured in a stationary frame with the origin at O, as

$y(x, t)=f(x - vt)$ (1)

Similarly, if the pulse travels to the left, the transverse positions of elements of the string are described by

$y(x, t) = f(x + vt)$ (2)

The function y, sometimes called **the wave function**, depends on the two variables x and t. For this reason, it is often written y(x, t)which is read "y as a function of x and t."

Furthermore, if t is fixed the wave function y(x), sometimes called the waveform, defines a curve representing the geometric shape of the pulse at that time

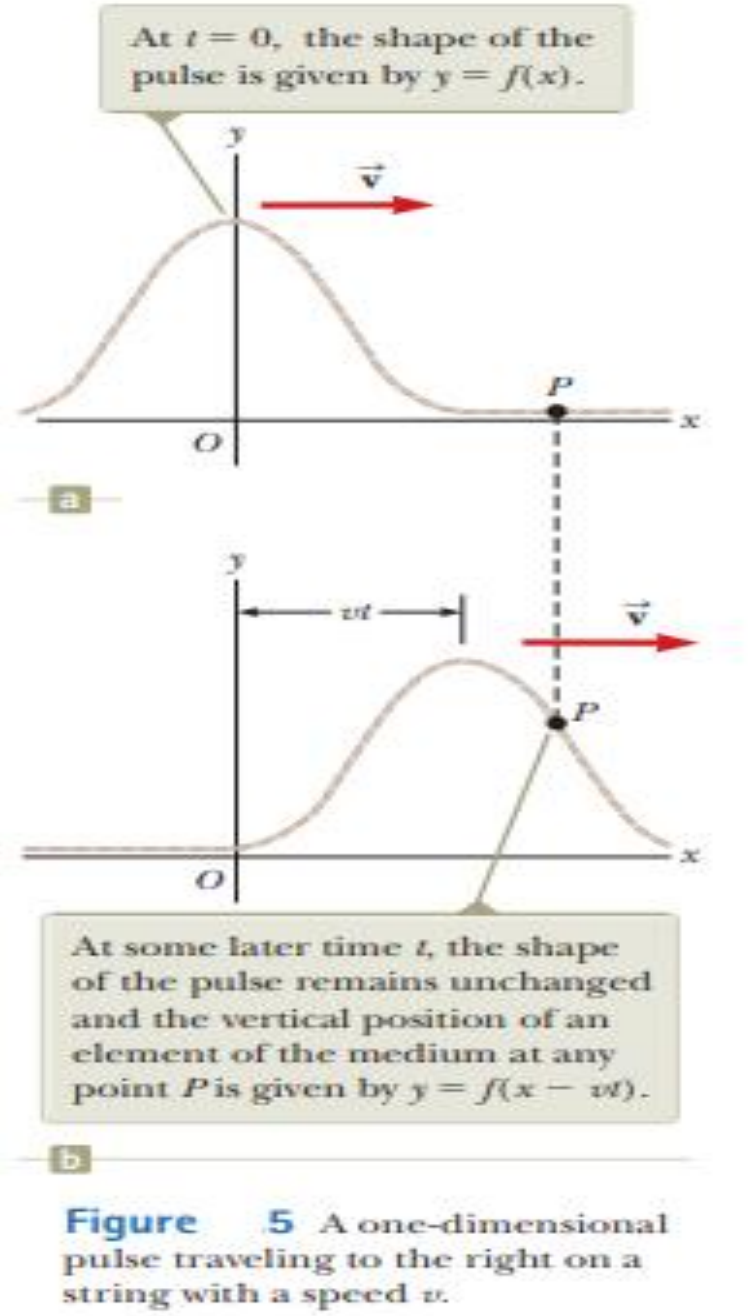


Figure 5 A one-dimensional pulse traveling to the right on a string with a speed v.

Example .1 A Pulse Moving to the Right

A pulse moving to the right along the x axis is represented by the wave function

$$y(x, t) = \frac{2}{(x - 3.0t)^2 + 1}$$

where x and y are measured in centimeters and t is measured in seconds. Find expressions for the wave function at $t = 0$, $t = 1.0$ s, and $t = 2.0$ s.

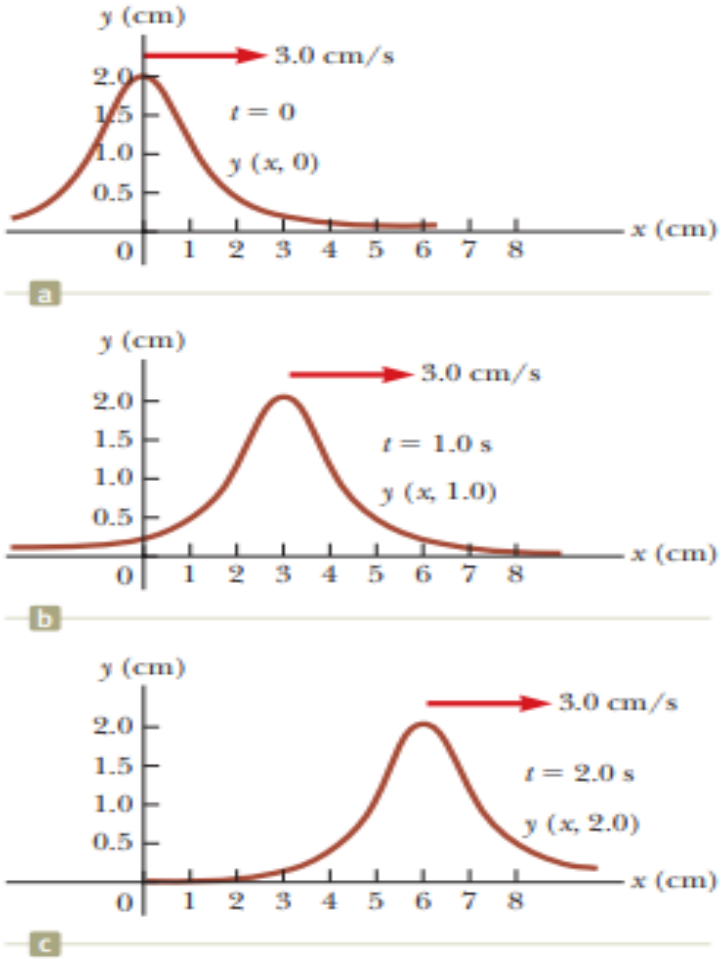


Figure .6 (Example 16.1) Graphs of the function $y(x, t) = 2/[(x - 3.0t)^2 + 1]$ at (a) $t = 0$, (b) $t = 1.0$ s, and (c) $t = 2.0$ s.

SOLUTION

Conceptualize Figure 6a shows the pulse represented by this wave function at $t = 0$. Imagine this pulse moving to the right at a speed of 3.0 cm/s and maintaining its shape as suggested by Figures .6b and .6c.

Categorize We categorize this example as a relatively simple analysis problem in which we interpret the mathematical representation of a pulse.

Analyze The wave function is of the form $y = f(x - vt)$. Inspection of the expression for $y(x, t)$ and comparison to Equation 1 reveal that the wave speed is $v = 3.0$ cm/s. Furthermore, by letting $x - 3.0t = 0$, we find that the maximum value of y is given by $A = 2.0$ cm.

Write the wave function expression at $t = 0$:

$$y(x, 0) = \frac{2}{x^2 + 1}$$

Write the wave function expression at $t = 1.0$ s:

$$y(x, 1.0) = \frac{2}{(x - 3.0)^2 + 1}$$

Write the wave function expression at $t = 2.0$ s:

$$y(x, 2.0) = \frac{2}{(x - 6.0)^2 + 1}$$

For each of these expressions, we can substitute various values of x and plot the wave function. This procedure yields the wave functions shown in the three parts of Figure 6.

Traveling Wave

- In this section, we introduce an important wave function whose shape is shown in Figure.7.
- The wave represented by this curve is called a sinusoidal wave because the curve is the same as that of the function sin Θ plotted against Θ.
- A sinusoidal wave could be established on the rope in Figure.1 by shaking the end of the rope up and down in simple harmonic motion.
- The sinusoidal wave is the simplest example of a periodic continuous wave and can be used to build more complex waves .
- The brown curve in Figure.7 represents a snapshot of a traveling sinusoidal wave at t = 0, and the blue curve represents a snapshot of the wave at some later time t. Imagine two types of motion that can occur. First, the entire waveform in Figure.7 moves to the right so that the brown curve moves toward the right and eventually reaches the position of the blue curve. This movement is the motion of the wave

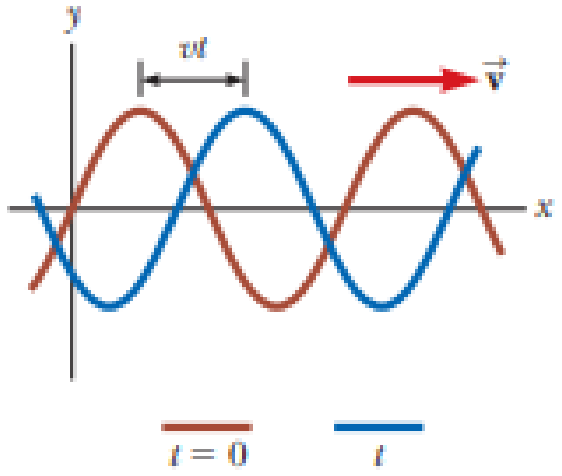


Figure .7 A one-dimensional sinusoidal wave traveling to the right with a speed v. The brown curve represents a snapshot of the wave at $t = 0$, and the blue curve represents a snapshot at some later time t.

- In what follows, we will develop the principal features and mathematical representations of the analysis model of a **traveling wave**. This model is used in situations in which a wave moves through space without interacting with other waves or particles.

Figure.8a shows a snapshot of a traveling wave moving through a medium. Figure 16.8b shows a graph of the position of one element of the medium as a function of time.

- A point in Figure.8a at which the displacement of the element from its normal position is highest is called **the crest** of the wave. The lowest point is called **the trough**.
- The distance from one crest to the next is called **the wavelength λ** (Greek letter lambda). More generally, the wavelength is the minimum distance between any two identical points on adjacent waves as shown in Figure .8a.
- If you count the number of seconds between the arrivals of two adjacent crests at a given point in space, you measure **the period T** of the waves. In general, the period is the time interval required for two identical points of adjacent waves to pass by a point as shown in Figure.8b.
- the inverse of the period, is called **the frequency *f***. In general, the frequency of a periodic wave is the number of crests (or troughs, or any other point on the wave) that pass a given point in a unit time interval. The frequency of a sinusoidal wave is related to the period by the expression :

$$f = \frac{1}{}$$

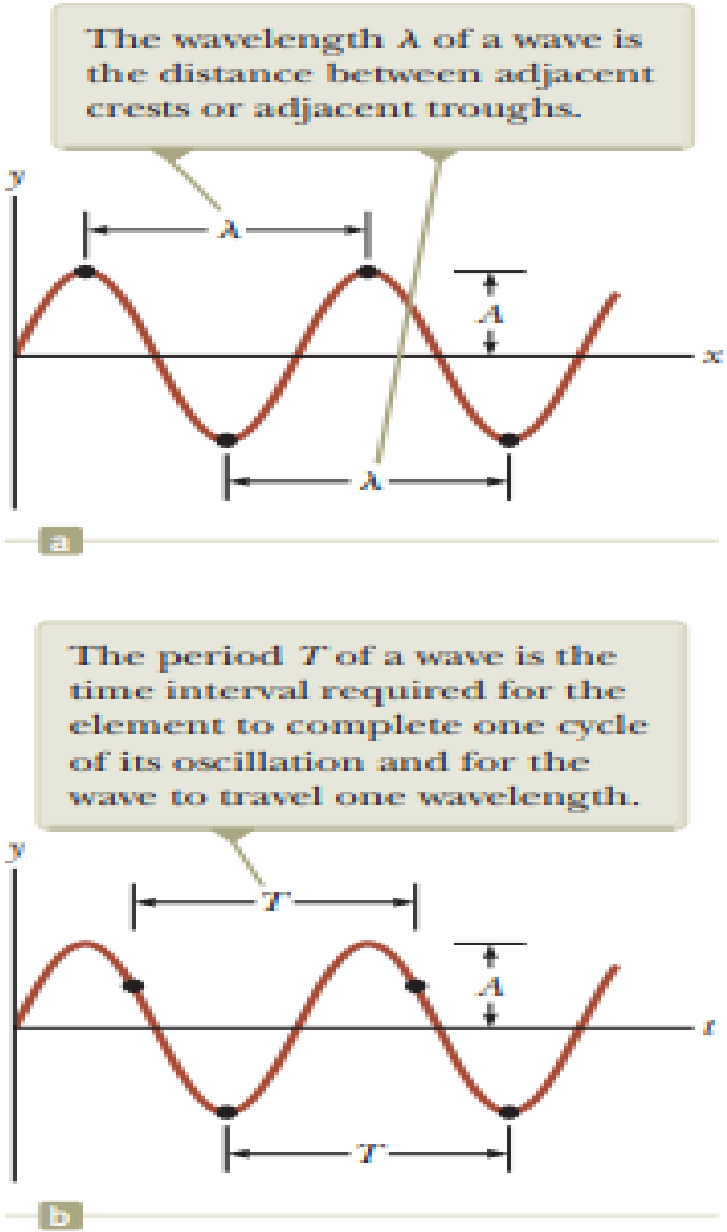


Figure .8 (a) A snapshot of a sinusoidal wave. (b) The position of one element of the medium as a function of time.

The most common unit for frequency, is s^{-1}, or hertz (Hz). The corresponding unit for T is seconds.

- The maximum position of an element of the medium relative to its equilibrium position is called the **amplitude** A of the wave as indicated in Figure 16.8

Wave function for a sinusoidal wave

Consider the sinusoidal wave in Figure .8a, which shows the position of the wave at t = 0.

Because the wave is sinusoidal, we expect the wave function at this instant to be expressed as

$$y(x, 0) = A \sin ax, \qquad (1)$$

where A is the amplitude and a is a constant to be determined.

At x = 0, we see that

$$y(0, 0) = A \sin a(0) = 0,$$

consistent with Figure 16.8a.

The next value of x for which y is zero is x = λ/2.

Therefore,

$$y\left(\frac{\lambda}{2}, 0\right) = A \sin\left(a\frac{\lambda}{2}\right) = 0$$

For this equation to be true, we must have aλ/2 = π, or a = 2π/λ.

Therefore, the function describing the positions of the elements of the medium through which the sinusoidal wave is traveling can be written

$$y(x, 0) = A \sin\left(\frac{2\pi}{\lambda}x\right) \qquad (2)$$

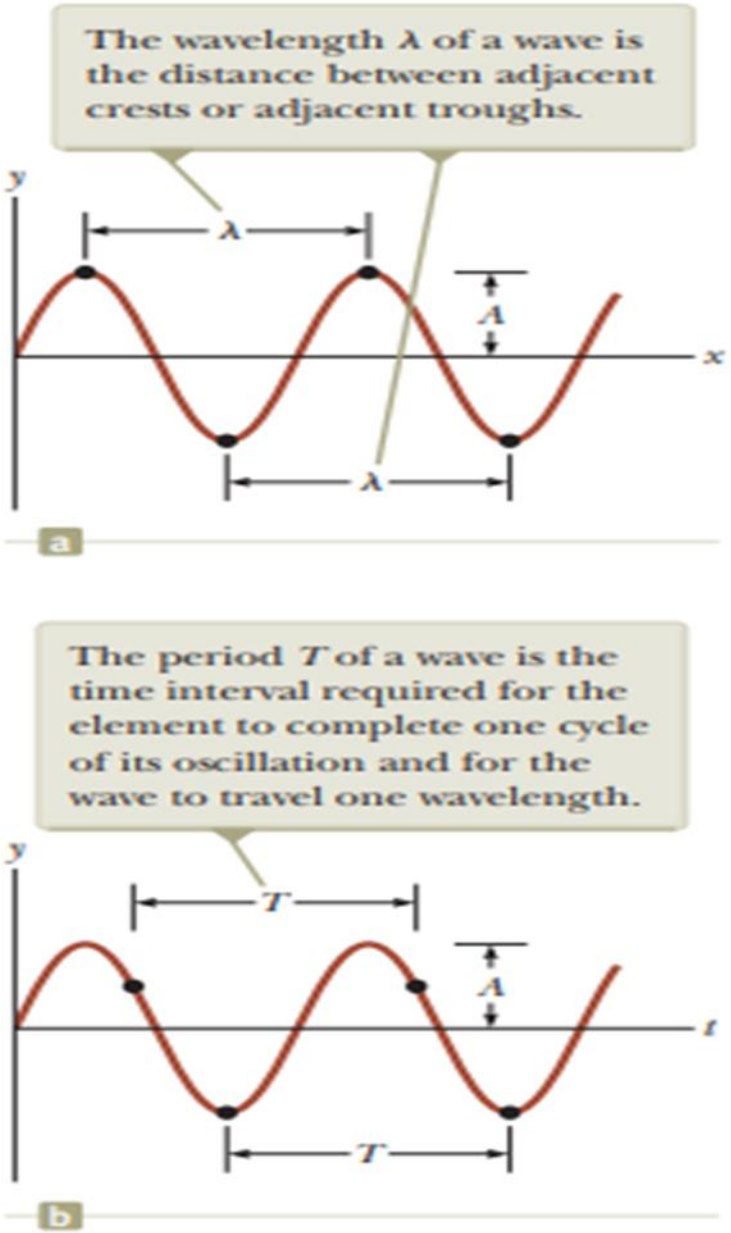


Figure .8 (a) A snapshot of a sinusoidal wave. (b) The position of one element of the medium as a function of time.

Based on our discussion of Equation **y(x, t)=f(x - vt)** **(1)**

if the wave moves to the right with a speed v, the wave function at some later time t is

$$y(x,t) = A\sin\left[\frac{2\pi}{\lambda}(x - vt)\right] \qquad (3)$$

By definition, the wave travels through a displacement Δx equal to one wavelength l in a time interval Δt of one period T. Therefore, the wave speed, wavelength, and period are related by the expressio

$$v = \frac{\Delta x}{\Delta t} = \frac{\lambda}{T}$$

Substituting this expression for v into Equation 3 gives

$$y = A\sin\left[2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right] \qquad (4)$$

We can express the wave function in a convenient form by defining two other quantities, the angular wave number k (usually called simply the wave number) and the angular frequency ω:

$$k \equiv \frac{2\pi}{\lambda}$$

$$\omega \equiv \frac{2\pi}{T} = 2\pi f$$

Substituting this expression into Equation 4 gives

$$y = A \sin (kx - \omega t) \qquad (5)$$

and this is the Wave function for a sinusoidal wave

the General expression for a sinusoidal wave

The wave function given by Equation 5 assumes the vertical position y of an element of the medium is zero at x = 0 and t = 0. That need not be the case. If it is not, we generally express the wave function in the form

$$y = A \sin (kx - \omega t + \phi)$$

where Φ is the phase constant. This constant can be determined from the initial conditions.

Example 16.2 A Traveling Sinusoidal Wave AM

A sinusoidal wave traveling in the positive x direction has an amplitude of 15.0 cm, a wavelength of 40.0 cm, and a frequency of 8.00 Hz. The vertical position of an element of the medium at $t = 0$ and $x = 0$ is also 15.0 cm as shown in Figure 16.9.

(A) Find the wave number k, period T, angular frequency ω, and speed v of the wave.

SOLUTION

Conceptualize Figure 16.9 shows the wave at $t = 0$. Imagine this wave moving to the right and maintaining its shape.

Categorize From the description in the problem statement, we see that we are analyzing a mechanical wave moving through a medium, so we categorize the problem with the *traveling wave* model.

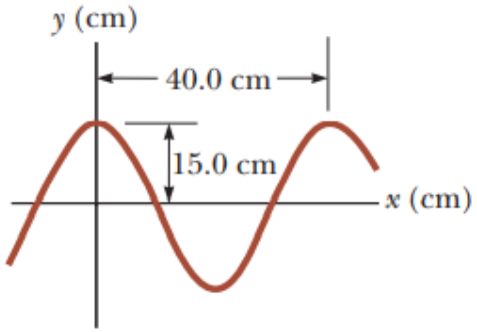


Figure 16.9 (Example 16.2) A sinusoidal wave of wavelength $\lambda = 40.0$ cm and amplitude $A = 15.0$ cm.

Analyze

Evaluate the wave number from Equation 16.8:

$$k = \frac{2\pi}{\lambda} = \frac{2\pi \text{ rad}}{40.0 \text{ cm}} = 15.7 \text{ rad/m}$$

Evaluate the period of the wave from Equation 16.3:

$$T = \frac{1}{f} = \frac{1}{8.00 \text{ s}^{-1}} = 0.125 \text{ s}$$

Evaluate the angular frequency of the wave from Equation 16.9:

$$\omega = 2\pi f = 2\pi(8.00 \text{ s}^{-1}) = 50.3 \text{ rad/s}$$

Evaluate the wave speed from Equation 16.12:

$$v = \lambda f = (40.0 \text{ cm})(8.00 \text{ s}^{-1}) = 3.20 \text{ m/s}$$

continued

(B) Determine the phase constant ϕ and write a general expression for the wave function.

SOLUTION

Substitute $A = 15.0$ cm, $y = 15.0$ cm, $x = 0$, and $t = 0$ into Equation 16.13:

$$15.0 = (15.0)\sin\phi \rightarrow \sin\phi = 1 \rightarrow \phi = \frac{\pi}{2}\text{ rad}$$

Write the wave function:

$$y = A\sin\left(kx - \omega t + \frac{\pi}{2}\right) = A\cos(kx - \omega t)$$

Substitute the values for A, k, and ω in SI units into this expression:

$$y = 0.150\cos(15.7x - 50.3t)$$

Reflection and Transmission

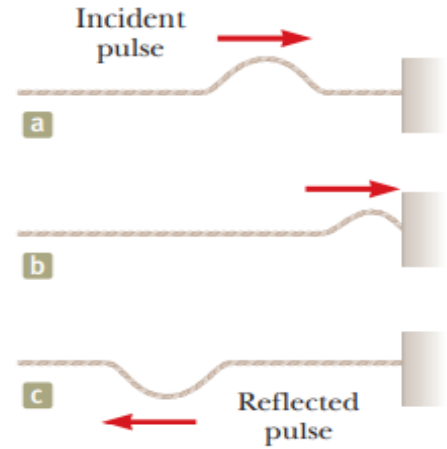


Figure 16.13 The reflection of a traveling pulse at the fixed end of a stretched string. The reflected pulse is inverted, but its shape is otherwise unchanged.

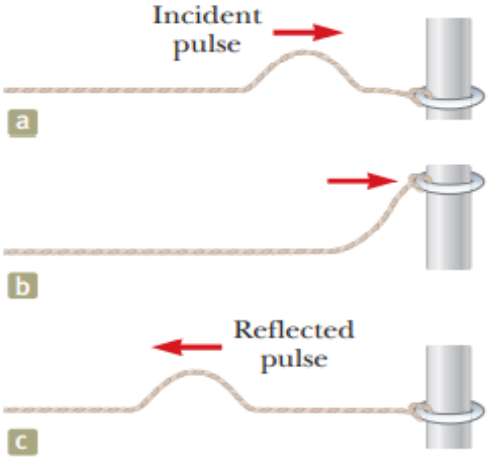


Figure 16.14 The reflection of a traveling pulse at the free end of a stretched string. The reflected pulse is not inverted.

The traveling wave model describes waves traveling through a uniform medium without interacting with anything along the way. We now consider how a traveling wave is affected when it encounters a change in the medium. For example, consider a pulse traveling on a string that is rigidly attached to a support at one end as in Figure 16.13. When the pulse reaches the support, a severe change in the medium occurs: the string ends. As a result, the pulse undergoes **reflection;** that is, the pulse moves back along the string in the opposite direction.

Notice that the reflected pulse is *inverted*. This inversion can be explained as follows. When the pulse reaches the fixed end of the string, the string produces an upward force on the support. By Newton's third law, the support must exert an equal-magnitude and oppositely directed (downward) reaction force on the string. This downward force causes the pulse to invert upon reflection.

Now consider another case. This time, the pulse arrives at the end of a string that is free to move vertically as in Figure 16.14. The tension at the free end is maintained because the string is tied to a ring of negligible mass that is free to slide vertically on a smooth post without friction. Again, the pulse is reflected, but this time it is not inverted. When it reaches the post, the pulse exerts a force on the free end of the string, causing the ring to accelerate upward. The ring rises as high as the incoming pulse, and then the downward component of the tension force pulls the ring back down. This movement of the ring produces a reflected pulse that is not inverted and that has the same amplitude as the incoming pulse.

Finally, consider a situation in which the boundary is intermediate between these two extremes. In this case, part of the energy in the incident pulse is reflected and part undergoes **transmission;** that is, some of the energy passes through the boundary. For instance, suppose a light string is attached to a heavier string as in Figure 16.15. When a pulse traveling on the light string reaches the boundary between the two strings, part of the pulse is reflected and inverted and part is transmitted to the heavier string. The reflected pulse is inverted for the same reasons described earlier in the case of the string rigidly attached to a support.

The reflected pulse has a smaller amplitude than the incident pulse. In Section 16.5, we show that the energy carried by a wave is related to its amplitude. According to the principle of conservation of energy, when the pulse breaks up into a reflected pulse and a transmitted pulse at the boundary, the sum of the energies of these two pulses must equal the energy of the incident pulse. Because the reflected pulse contains only part of the energy of the incident pulse, its amplitude must be smaller.

When a pulse traveling on a heavy string strikes the boundary between the heavy string and a lighter one as in Figure 16.16, again part is reflected and part is transmitted. In this case, the reflected pulse is not inverted.

In either case, the relative heights of the reflected and transmitted pulses depend on the relative densities of the two strings. If the strings are identical, there is no discontinuity at the boundary and no reflection takes place

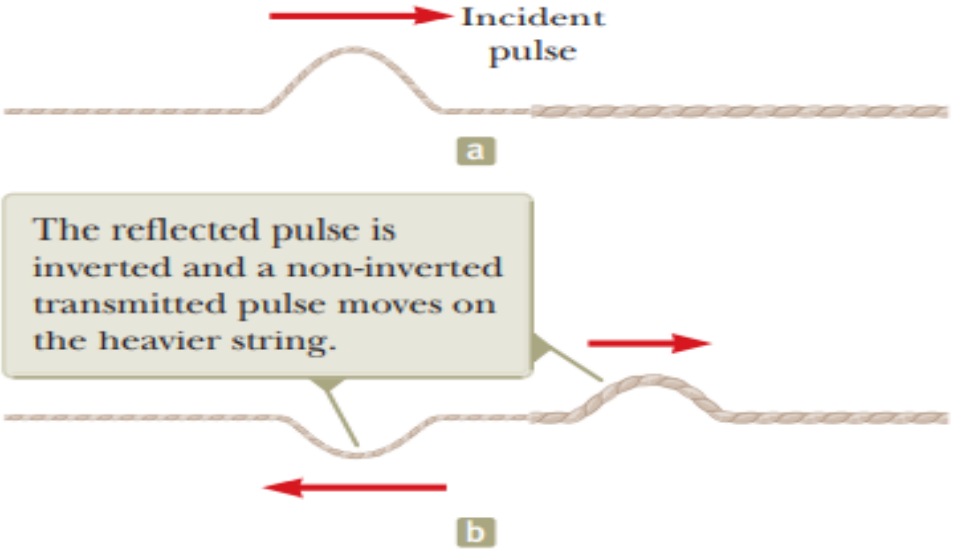


Figure 16.15 (a) A pulse traveling to the right on a light string approaches the junction with a heavier string. (b) The situation after the pulse reaches the junction.

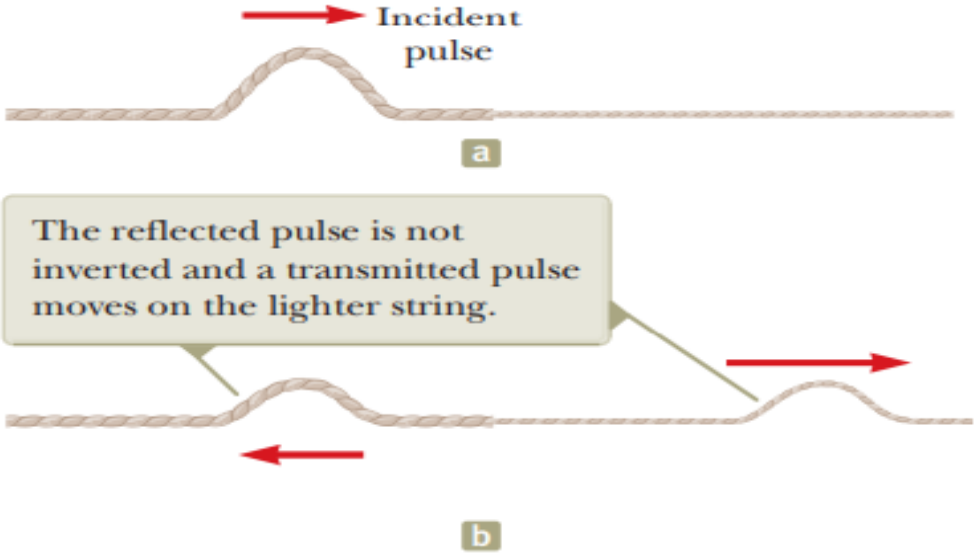


Figure 16.16 (a) A pulse traveling to the right on a heavy string approaches the junction with a lighter string. (b) The situation after the pulse reaches the junction.

According to Equation 16.18, the speed of a wave on a string increases as the mass per unit length of the string decreases. In other words, a wave travels more rapidly on a light string than on a heavy string if both are under the same tension. The following general rules apply to reflected waves: When a wave or pulse travels from medium A to medium B and $v_A > v_B$ (that is, when B is denser than A), it is inverted upon reflection. When a wave or pulse travels from medium A to medium B and $v_A < v_B$ (that is, when A is denser than B), it is not inverted upon reflection.